MAKERERE UNIVERSITY

COLLEGE OF COMPUTING & INFORMATION SCIENCES

SCHOOL OF COMPUTING & INFORMATICS TECHNOLOGY

CSC2107 DATABASE MANAGEMENT SYSTEMS
TEST 2 (Computer Science Year-2 A) 2022/2023

TIME ALLOWED: 1 Hour 15 Minutes

Instructions

1. Attempt all questions

Question One

A company called *FastCab* provides a taxi service to clients. The table below represents the First Normal Form (1NF) for a client bookings for taxis. Study it and answer the following questions.

a) Identify the functional dependencies that exist between the columns in the table. *(5 Marks)*

b) Describe and illustrate the process of normalizing the attributes shown in the table to produce a set of well-designed 3NF (*Third Normal Form*) Relations. *(15 Marks)*

c) Identify the primary, alternate and foreign keys in your 3NF relations. *(5 Marks)*

d) Identify any relationships between the 3NF relations in (b) above *(5 Marks)*

e) Create an Entity Relationship (E-R) Model to represent the data stored by FastCab company based on the 3NF relations in (b) and the relationships in (d) above.

(10 Marks)

JobID	JobDate Time	driverID	driver Name	taxiID	clientID	clientName	jobPickUpAddress
1	25/07/14 10.00	D1	Joe Bull	T1	C1	Anne Woo	1 Storrie Rd, Paisley
2	29/07/14 10.00	D1	Joe Bull	T1	C1	Anne Woo	1 Storrie Rd, Paisley
3	30/07/14 11.00	D2	Tom Win	T2	C1	Anne Woo	3 High Street, Paisley
4	2/08/14 13.00	D3	Jim Jones	T3	C2	Mark Tin	1A Lady Lane, Paisley
5	2/08/14 13.00	D4	Steven Win	T1	C3	John Seal	22 Red Road, Paisley
6	25/08/14 10.00	D2	Tom Win.	T2	C4	Karen Bow	17 High Street, Paisley

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